The Australian honey bee industry is composed of about 29,690 registered beekeepers. Around 2000 of these are considered to be commercial apiarists, each with more than 50 hives (average 400 – 800 hives).

The European honey bee contributes directly to the Australian economy. The industry produces about 37,000 tonnes of honey per annum. This is the major output of the industry. Other sources of income include paid pollination services, beeswax production, and queen bee and packaged bee sales. In 2021, the value of honey bee hive products was estimated at $224 million per annum.

Honey bees also contribute to the productivity of many horticultural and seed crops, by providing essential pollination services that improve crop yield and quality. The unrecognised value of pollination services have been estimated to be in the order of $14.2 billion per annum.

Honey bee pests and diseases

European honey bees may be affected by a range of pests and diseases. Some of these are exotic to Australia.

Exotic bees, bee pests and diseases

- Acute bee paralysis virus
- African honey bee (Apis mellifera scutellata)
- Africanised honey bee (hybrid Apis mellifera scutellata)
- Cape honey bee (Apis mellifera capensis)
- Deformed wing virus
- Hornets (exotic Vespa spp.)
- Large hive beetle (Oplostomus fuligineus)
- Slow paralysis virus
- Tropilaelaps mite (Tropilaelaps clareae)
- Tracheal mite (Acarapis woodi)
- Varroa jacobsoni mite

Some of these pests and diseases are already established in Australia

- American foulbrood (bacteria, Paenibacillus larvae)
- Black queen cell virus (virus, Cripavirus genus)
- Braula fly (Braula caeca)
- Chalkbrood disease (Ascosphaera apis)
- European foulbrood (Melissococus pluton)
- Leafcutter bee chalkbrood (Ascosphaera aggregata)
- Nosema (Nosema apis and Nosema ceranae)
- Sacbrood virus (virus, Iflavirus)
- Small hive beetle (Aethina tumida)
- Stonebrood (Aspergillus falvum and A. fumigatus)

- Varroa mite (Varroa destructor)
- Wax moth (Galleria mellonella and Achroia grisella)

Further information on established and exotic pests and diseases of European honey bees is available on the BeeAware and Plant Health Australia websites. For current information about varroa mite management go to the Australian Honey Bee Industry Council (AHBIC) website.

Preventing the entry of exotic pests and diseases

Our department undertakes a range of activities to try to prevent the entry of exotic bees, bee pests and diseases including:

- surveillance outside Australia through the International Plant Health Surveillance Program, with our staff developing and implementing measures for the early detection of targeted pests and diseases
- inspections of people, mail parcels, baggage, ships, animals, plants and cargo containers by our staff to help prevent the entry into Australia of foreign bees and any pests and diseases they carry
- surveillance within Australia to detect any incursions to enable the pest or disease to be destroyed before it becomes established or restrict its spread.

Detecting incursions of exotic pests and diseases

The National Bee Pest Surveillance Program (part of the National Bee Biosecurity Program) is an early warning system to detect new incursions of exotic bee pests and pest bees. The program involves a range of surveillance methods conducted at locations considered to be the most likely entry point of bee pests and pest bees throughout Australia. Plant Health Australia (PHA), at a national level, coordinates and administers the program, which is jointly funded by the Australian Honey Bee Industry Council (AHBIC), Horticulture Innovation Australia Ltd, Grain Producers Australia and the Australian Government through the Department of Agriculture, Fisheries and Forestry.

Improving the management of established pests and diseases

AHBIC and PHA have worked to establish a National Bee Biosecurity Program. The purpose of this program is to improve the management of established pests and diseases, as well as increase the preparedness and surveillance of exotic pest threats of the honey bee industry. The Program will be underpinned by a beekeeping Code of Practice, a Bee Biosecurity Officer in each state to conduct inspections, as well as educate and train beekeepers in biosecurity best practice. Through a statutory industry biosecurity levy paid by bee keepers, the industry is investing around $400,000 per annum in the program.

Protecting your honey bees

If exotic bees, or honey bee pests and diseases enter Australia, early detection will be crucial in limiting their spread and impact on the Australian honey bee industry. All beekeepers (including commercial and backyard beekeepers) have a significant role in recognising and reporting any suspected infestation of exotic pests and diseases (or other pests and diseases).

Know what to look for. If you suspect the presence of exotic bees, bee pest or diseases, immediately:

- phone the Exotic Plant Pest Hotline: free call 1800 084 881
- implement appropriate controls
- be aware of your biosecurity obligations – Australian Honey Bee Industry Biosecurity Code of Practice
- participate in the annual BeeBlitz month
- follow any biosecurity instructions in the event of an incursion
- don't try to bypass biosecurity controls aimed at protecting the industry.

Research and development

The agriculture portfolio's investment in research and development for the honey bee industry is managed by AgriFutures. AgriFutures commissions a range of research projects aimed at improving the management of established bee pests and diseases and improving the prospects of detecting and eradicating exotic pests and diseases before they establish.

More information, including research currently underway and reports from completed projects, is available on AgriFutures's Honey Bee and Pollination Program website.

In June 2016 Hort Innovation launched a significant pollination research investment fund to increase crop quality and yields through more effective pollination and alternate pollinators. Supported with Australian Government funding, the fund comprises multiple projects being delivered in partnership with co-investors such as research institutions, government agencies or international and commercial enterprises. More information about research currently underway is available on Hort Innovation's Pollination Fund webpage.

Scientists at the Commonwealth Scientific and Industrial Research Organisation (CSIRO) are playing a role in global research networks to better understand the causes of declining bee health in many parts of the world. More recent innovation by CSIRO scientists has led to enormous improvement in microsensor technology for tracking bees in and around hives and the opportunity for new insights into hive health and bee response to diseases or chemicals. More information on this work is available from the CSIRO website.

A range of research groups based at Australian universities are also studying honey bee health. Some of these research groups are:

- The Centre for Integrative Bee Research at the University of Western Australia

- The Social Insects Lab at the University of Sydney
- Plants, Animals and Interactions, University of Western Sydney

Many of Australia's northern neighbours harbour major mite pests—Tropilaelaps clareae and Varroa jacobsoni —that could cause significant damage to our honey bee population. The Australian Centre for International Agricultural Research (ACIAR) has funded research, with CSIRO as the commissioned organisation, on these pests for about 15 years, making a significant contribution to a better understanding of the mites and their host conditions.

These projects, carried out in Australia, Indonesia, Papua New Guinea and the Philippines, have substantially increased understanding about these pests, to the extent that the picture of their spread and the risks they pose is better understood. There are two major benefits from the research: to beekeeping through better understanding of mite control methods; and to biosecurity procedures through better understanding of the true nature of the risks posed by the mites.

Agricultural chemical regulation

The Australian Pesticides and Veterinary Medicines Authority (APVMA) is the statutory agency responsible for assessing and registering agricultural chemicals for use in Australia.

The APVMA is aware of concerns that insecticides, especially those of the neonicotinoid class, may be contributing to a decline in honey bee populations in Europe and the United States. The APVMA has completed a report (Overview report – Neonicotinoids and the health of honey bees in Australia) on the issues relating to honey bee health in Australia, with a particular focus on the use of neonicotinoid insecticides.

The APVMA's report concluded that there is lack of consensus in the scientific literature on the causes of regional declines in honey bee populations in Europe and the United States. A wide range of possible causes are being actively investigated including pesticides, parasites, viruses, climate change, bee nutrition, lack of genetic diversity, and beekeeping practices.

Australia is working internationally through the OECD Pesticides and Pollinators Working Group and the Society of Environmental Toxicology and Chemistry to develop expanded test methods and guidelines for assessing the effect of pesticides on insect pollinators. These expanded methods assess the possible sub-lethal effects of some pesticides on honey bees at very low concentrations (one part per billion and less).

Further information

The Australian Honey Bee Industry Council (AHBIC) is the national representative body for the honey bee industry. Their website is https://honeybee.org.au.

The BeePestBlitz website contains information about the annual Bee Pest Blitz campaign. The campaign aims to create awareness of exotic and established bee pests, the importance of hive inspections using nationally-agreed surveillance techniques and consistent record keeping and reporting of results.

The BeeAware website is a source of information for beekeepers and growers about honey bee biosecurity and pollination of agricultural and horticultural crops. This website has not been actively maintained since 2022.

BeeAware contains an extensive range of information about exotic and established pests and diseases of honey bees and helps beekeepers to identify and respond to these pest threats. It also contains information about the pollination of crops and how beekeepers and growers can work together to provide and receive best practice pollination services.

BeeConnected is a user-driven smart-phone app that enables collaboration between beekeepers, farmers and spray service contractors to facilitate best-practice pollinator protection. It was developed by Croplife Australia in partnership with the Australian Honey Bee Industry Council.